

M002 Extension 2 — Worksheet (Beginner)

Topic: Two Pistons + AND Conditions · **Course:** Maze Madness · **Level:** Extension (Beginner) · **Time:** about 35 minutes

This is a **challenge** for students who finished the weekly mazes. The agent follows a **redstone trail**. Sometimes it must check **two things at once** (this is called **AND**). Twice in the maze it flips a **lever** to power a **piston** that opens the way. Your goal: get the agent through the **whole maze**.

1 · Read the Chat Commands

A chat command binds a **word** to an action. When you type that word in chat, the agent does it.

```
when you type "r":
    agent turns right
```

You type **r**. What does the agent do? Circle one: turns left · turns right · jumps


```
when you type "r1":
    agent teleports back to you
```

You type **r1**. Where does the agent go? Circle one: to the goal · back to you · nowhere

2 · AND means BOTH 🔍

AND is true only when **both** things are true. Read each line and answer.

```
if redstone on left AND redstone ahead:  
    turn left
```

There is redstone on the left, but NOT ahead. Does the agent turn left? Circle one: yes · no

There is redstone on the left AND redstone ahead. Does the agent turn left? Circle one: yes · no


```
if block ahead:  
    flip the lever  
    move forward
```

A block is in the way (a piston gate). What does the agent do first? Circle one: flip the lever · move back · stop

3 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version. Circle what's different and write one short sentence about the bug.

Pair A — The agent should turn left only when there is redstone on the left **AND** ahead.

```
# clean
if redstone on left AND redstone ahead:
    turn left
```

```
# buggy
if redstone on left OR redstone ahead:
    turn left
```

What is wrong? With OR, how many of the two does the agent need?

Pair B — At a piston gate the agent should flip the lever **first**, then cross.

```
# clean
flip the lever
move forward
move forward
```

```
# buggy
move forward
move forward
flip the lever
```

What is wrong? What does the agent hit if the lever is not flipped yet?

4 · Fill the Gap

The agent should turn left only when **both** are true. One word is missing. Fill it in from the word bank.

```
if redstone on left ____ redstone ahead:  
    turn left
```

Word bank: AND · OR · NOT

Write the missing word:

run

x here

e age

needs

6 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while the agent solves the whole maze. Talk like you are teaching a friend. Try to use these words: **redstone, AND, piston, lever.**

1. Show the maze and the redstone trail.
2. Type **run** and show the agent following it.
3. Show the agent flipping a lever to open a piston.
4. Show it reaching the goal.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.